

Proof: $MSE(\hat{f}, f) = B^2 + V$

with $B = E[\hat{f}(x)] - f(x)$; $V = \text{var}[\hat{f}(x)]$.

Proof: start with:

$$MSE(\hat{f}, f) = E[(f - \hat{f})^2]$$

$$= E[f^2 - 2f\hat{f} + \hat{f}^2]$$

$$= f^2 - 2fE[\hat{f}] + E[\hat{f}^2]$$

adding
in
zero

$$= E[\hat{f}]^2 - E[\hat{f}]^2 + f^2 - 2fE[\hat{f}] + E[\hat{f}^2]$$

$$= E[\hat{f}]^2 - 2E[\hat{f}]f + f^2 + E[\hat{f}^2] - E[\hat{f}]^2$$

$$= \underbrace{(E[\hat{f}] - f)^2}_{\text{Bias!}} - \underbrace{E[\hat{f}^2] - E[\hat{f}]^2}_{\text{Variance!}}$$

$$= B^2 + V \quad \text{as desired}$$

